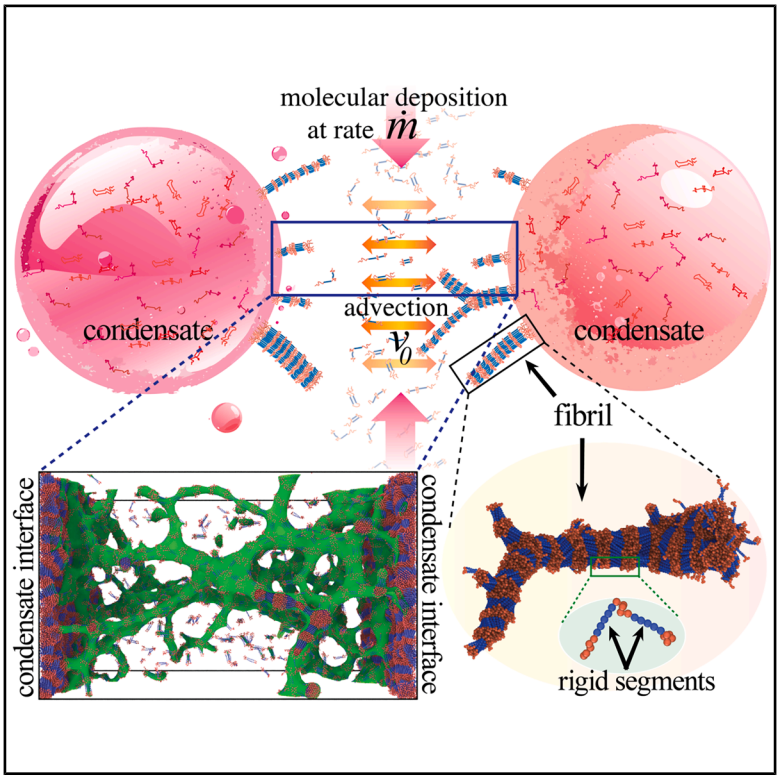
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Article

β motifs and molecular flux promote fibril growth at condensate interfaces

Graphical abstract



Authors

Subhadip Biswas, Davit A. Potoyan

Correspondence

subhadip.biswas@cnrs.fr (S.B.),
potoyan@iastate.edu (D.A.P.)

In brief

Biomolecules; Chemical reaction kinetics; Flux data

Highlights

- Rigid β-prone segments convert interfacial adsorption into protrusive fibril growth
- Molecular deposition controls whether chains spread laterally or extend outward
- Higher initial drift speeds flat surface growth but slows axial fibril elongation
- Interfacial orientational order couples molecular architecture to condensate aging

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β motifs and molecular flux promote fibril growth at condensate interfaces

Subhadip Biswas[1,2,*] and Davit A. Potoyan[1,3,4,*]
[1]Department of Chemistry, Iowa State University, Ames, IA 50011, USA
[2]Institut Charles Sadron, Université de Strasbourg & CNRS, 23 rue du Loess, 67034 Strasbourg, Cedex, France
[3]Department of Biochemistry, Biophysics and Molecular Biology, Iowa State University, Ames, IA 50011, USA
[4]Lead contact
*Correspondence: subhadip.biswas@cnrs.fr (S.B.), potoyan@iastate.edu (D.A.P.)
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SUMMARY

Biomolecular condensates can age into gel-like or solid-like states, and their interfaces can promote amyloid fibril formation. Yet the physical principles selecting distinct interfacial aging pathways remain unclear. Here, we develop flux-driven molecular dynamics (FD-MD), a coarse-grained framework to disentangle the roles of rigid β-prone motifs, hydrophobic sequence patterning and molecular deposition rate. We show that rigid motifs can sustain orientational order that converts interfacial adsorption into directional fibrillar growth, whereas flexible chains form spreading surface aggregates. Furthermore for fixed molecular architecture, changing the rate and manner of molecular delivery redirects assembly between lateral spreading, outward fibrillar growth, and bridging between opposing interfaces. Counterinutively, stronger initial drift toward the interface accelerates surface coverage but slows axial fibril elongation, consistent with competition between interfacial accumulation, orientational ordering and outward growth. Molecular architecture thus provides the capacity for directional assembly, while molecular supply selects it morphological outcome. Our findings provide a minimal physical framework for distinct condensate-aging pathwasy and suggest routes to redirect aging through molecular delivery and tuning of interfacial properties of condensates.

INTRODUCTION

Biomolecular condensates play key regulatory roles in cells by forming dynamic membraneless compartments that selectively partition, inhibit, or catalyze biochemical processes.[1–4] In their functional states, condensates typically display liquid-like behavior: they fuse, rearrange internally, and exchange molecules with their surroundings.[3,5,6] However, recent experiments reveal that this liquid-like material state can change with time,[7–9] in a process known as aging. During aging, condensates can lose their liquid-like state and, over time, evolve into viscoelastic, gel-like, glassy, or solid-like forms, often accompanied by the emergence of amyloid-like structures.[7,10–14] Understanding how dynamic condensate states transition to an arrested or fibrillar state is therefore a central problem linking condensate internal structures and material states to amylodiogenesis and age-related diseases.[4,13,15,16] The link between condensate aging and amyloid formation is being actively studied to understand causal relationships and the role of condensates in the early initiation phase of amyloid phase.

Recent experiments have shown that condensate interfaces can promote heterogeneous fibril nucleation, although this behavior is not universal across condensate systems.[17] In hnRNPA1 low-complexity-domain condensates and FUS condensates, fibril formation is strongly enhanced at the droplet surface.[9,18–20] In other systems, condensates can modulate fibrillation through interior sequestration, altered viscoelasticity, or phase-dependent changes in the aggregation pathway.[8,17,21] Condensates can also alter the structural polymorphism or spatial propagation of amyloid-forming proteins such as Tau and Aβ.[22–24] Thus, condensates do not simply accelerate aggregation by concentrating proteins. They can promote, redirect, suppress, or spatially localize fibrillation depending on metastability, composition, interfacial structure, and molecular exchange kinetics.[13,17,21,25] What governs this selection remains poorly understood. To better understand the role of condensate interfaces in amyloid formation, we need to separate molecular and interfacial determinants of condensate aging. Molecular determinants include sequence effects such as valence, sticker strength, spacing, and chain rigidity, all of which have been shown to influence condensate material states and morphologies.[26–31] In prion-like and low-complexity domains, the same interactions that stabilize reversible condensates can also expose or enrich motifs capable of forming β-sheet-rich assemblies.[32–34] The role of the condensate interface is more active than previously thought, and recent simulations and experiments indicate that protein

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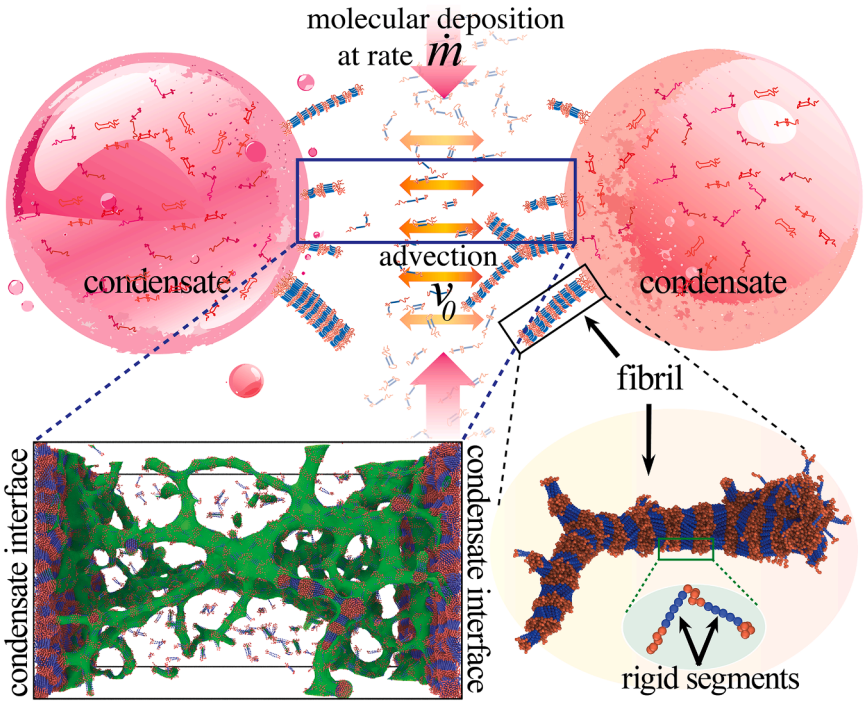


Figure 1. Conceptual overview of the FD-MD model

The schematic shows two adjacent spherical nascent condensates formed by single protein species. The same chains are introduced into the dilute region at a prescribed molecular deposition rate, $\dot{m}$. Each incoming chain is directed toward the nearest condensate interface with an initial drift velocity of magnitude $|v_0|$. The zoomed-in box between two opposing condensate surfaces defines the slab region used for the fibril-growth simulations. The simulated slab therefore contains two condensate interfaces, as shown in the left inset. The right inset illustrates fibril growth emerging from the condensate surface, and the magnified filament highlights the constituent chain molecules used in the simulations.

chains can adopt position-dependent conformations and orientations there, with scaffold and adsorbed components showing distinct interfacial anisotropies.[35–38] These effects are expected to depend on thermodynamic state and may weaken near criticality where interfacial contrast is reduced.[39,40] Recent work has distinguished homogeneous fibril nucleation in the dilute phase from heterogeneous nucleation promoted by condensate interfaces.[16,21] In the RIPK3 RHIM domain, the effect of condensation on fibrillation can switch with protein conformational state, suppressing fibril formation for predominantly folded proteins while promoting interface-mediated nucleation when the proteins are predominantly unfolded.[41] In A1-LCD systems, separate saturation concentrations for condensation and fibril formation show that condensates can remain metastable with respect to the fibrillar state; under these conditions, the dense interior can sequester protein, whereas the interface provides a distinct environment that promotes fibril formation.[21] Consistent with this picture, experiments on hnRNPA1 condensates have directly shown preferential fibril formation at the condensate interface,[9] and related forms of heterogeneous amyloid nucleation have been discussed at other soft interfaces, including lipid membranes.[13,42] In the present work, we therefore distinguish the experimentally established concept of interface-mediated heterogeneous nucleation from what is directly resolved in our simulations. Because the FD-MD model does not define a critical nucleus or measure nucleation rates or barriers, we use the term "interfacial fibril initiation" for the emergence of ordered, protrusive fibrillar structures at the condensate boundary, without assigning these events to a primary or secondary nucleation pathway.

Finally, while most experiments are done *in vitro*, we should acknowledge that in cells, condensates sit within a driven environment whose composition is maintained by synthesis, degradation, active transport, chemical remodeling, and exchange with surrounding phases.[2,40,43,44] Therefore, the rate at which aggregation-prone proteins and peptides reach an interface is not merely a background condition; it is itself a control parameter. Recent experiments on flow-coupled aging of ATP condensates illustrate how internal or external material flux can regulate condensate maturation.[45,46] Yet most models of condensate aging and fibrillization treat molecular exchange as either equilibrated or implicit, leaving unresolved how the deposition rate of new material controls the structure built at the interface. Coarse-grained and multiscale protein models provide sequence-level or chemically specific insight into aggregation and phase behavior.[47–49] Minimal amyloid simulations capture nucleation, elongation, nonspecific association, and fibril self-replication.[50–52] Mesoscale condensate models describe aging, desolvation, rigidification, viscoelasticity, and energy-landscape effects.[40,53–56] Concepts from polymer and glass physics further motivate the idea that arrested growth can emerge from the competition between local mobility, stress relaxation, and structural ordering.[57–60] However, the direct coupling between molecular grammar, deposition rate, initial drift velocity, and the resulting choice between spreading, arrest, protrusion, and bridging remains largely unresolved.

Here, we address this gap by introducing flux-driven molecular dynamics (FD-MD), a non-equilibrium coarse-grained framework in which chain architecture, chain rigidity, and molecular transport can be varied independently in the presence of condensate interfaces (Figure 1). In the model, the deposition rate $\dot{m}$ controls how frequently chains are introduced into the dilute region, while the initial drift velocity $|v_0|$ controls the magnitude of the initial velocity directed toward the condensate interface. Without an initial kick toward the interface, inserted molecules diffuse in the dilute phase and aggregate. Therefore, an initial drift velocity is needed to mimic the chemical-potential gradient that drags molecules toward the condensate interface, where they nucleate at the condensate corona. Together, $\dot{m}$ and $|v_0|$ define a minimal flux protocol for probing how non-equilibrium delivery shapes interfacial assembly of fiber-like structures. Using FD-MD, we find that

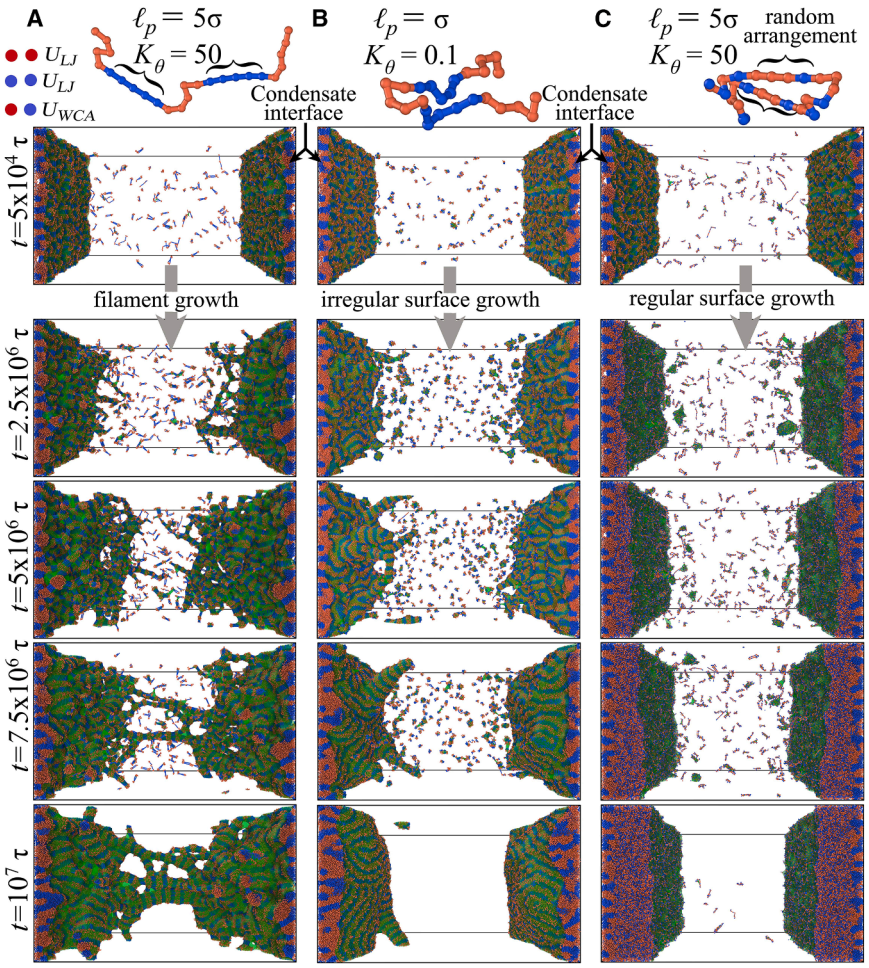


Figure 2. Simulation snapshots of FD-MD filament growth, irregular surface growth, and regular surface growth at biomolecular condensate interfaces

Two slabs of arrested chains present the interfaces of two adjacent condensates. New chains are added at the center of the dilute phase between two interfaces at a constant rate with an initial drift until N_p = 20, 000 chains have been introduced. The simulations illustrate the role of sequence-encoded rigidity in fibril formation.

(A) Polymers with two rigid blue β-prone sticker segments (persistence length $\ell_p = 5\sigma$) and floppy red linker segments nucleate at the condensate interface and progressively assemble into surface-anchored fibrillar protrusions. Over the interval shown (5 $\times$ 10^4–10^7 τ), these filaments elongate, bundle, and form mesh-like structures that can bridge opposing interfaces.

(B) When both the blue and the red segments are flexible, interfacial aggregation persists, but no fibrillar structures emerge.

(C) When the bead sequence is scrambled at fixed composition while the rigid segments are held at the same backbone positions as in (A), the chains fail to form filaments and instead promote the growth of extended planar interfaces. This image also removes the blue-blue attraction, so it changes both the sequence and the interaction matrix (see parameters in Table S1). See also Figure S5; Table S1.

fibrillar aging is controlled by sequence-encoded rigidity, interfacial geometry, and molecular flux. Chain rigidity determines whether molecules deposited onto the condensate surfaces can sustain ordered protrusions, while the deposition rate controls interfacial accumulation. The initial drift plays a more subtle role: stronger initial velocity can enhance surface coverage but can also suppress out-of-plane elongation by redistributing chains within the interfacial plane. As a result, flux acts as a switch between planar spreading and protrusive fibril growth. We further show that the condensate interface imposes a layered orientational order on incoming chains: arriving β-prone segments first lie flat in the interfacial plane, while segments displaced roughly one rod length above that contact layer stand along the interface normal. This is the arrangement a protrusion needs in order to elongate. Together, these results provide a physical mechanism linking molecular architecture, condensate interfaces, and non-equilibrium material supply to the pathways by which condensates age into fibrillar assemblies.

RESULTS

We first describe the model and simulation geometry, which are sketched in Figure 1; full simulation parameters and protocols are given in STAR Methods. The system consists of two opposing semi-arrested condensate surfaces located near the low- and high-x boundaries of a three-dimensional rectangular simulation box. These surfaces are constructed from 8,000 chains of the same molecular species that is later supplied to the dilute region to drive fibril growth. Thus, the pre-formed condensate surfaces and the incoming fibril-forming material have identical molecular composition; only their initial state and transport protocol differ. At the center of the dilute region (implicit solvent) between these surfaces, we introduce up to 20,000 additional chains at a prescribed deposition rate $\dot{m}$. We assign the incoming chains an initial velocity randomly, either with positive or negative drift toward the interface, allowing us to control molecular arrival independently of molecular architecture. Each chain is represented as a 25-bead coarse-grained polymer with two bead types, analogous to a minimal block-copolymer model. Homotypic red-red and blue-blue contacts interact attractively through Lennard-Jones potentials, whereas heterotypic red-blue contacts are purely repulsive and are modeled using a Weeks-Chandler-Andersen (WCA) potential (Figure 1). The patterned chains contain the same overall bead composition, but differ in the spatial arrangement of cohesive motifs along the backbone (Figure 2C). Thus, changes in patterning alter the spacing and placement of aggregation-prone segments without changing the total amount of cohesive material.

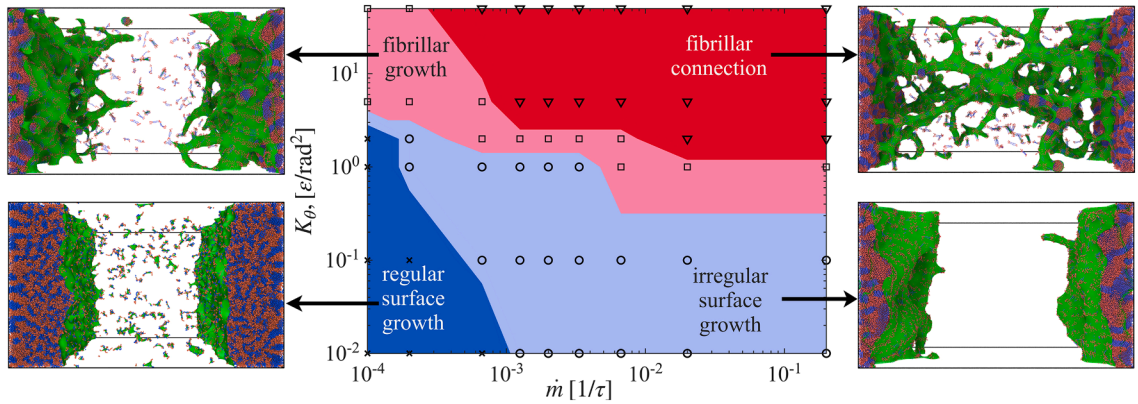


Figure 3. Interfacial fibril growth morphologies as a function of chain stiffness

K_θ and molecular deposition $\dot{m} = 1/\tau_d$, with the initial drift given by the potential in Equation 2. Phase diagram obtained from FD-MD simulations under mass flux toward a condensate interface, shown as a function of deposition rate $\dot{m}$ ($\tau_d = 1/\dot{m}$) and bending stiffness (or persistence length) of the β-prone segment. Four distinct growth regimes are observed: (I) planar surface growth or complete wetting of the condensate surface (fully flexible chains) characterized by smooth interfacial thickening; (II) irregular surface deposition at low but finite stiffness, where disordered aggregates accumulate without long-range order; (III) fibrillar interfacial growth at intermediate-to-high stiffness (partially wetted surface), marked by nucleation and elongation of surface-anchored fibrils; and (IV) bridging and network-resulting structures at high stiffness and high deposition rate, where fibrils bundle and connect opposing interfaces. Symbols give the end-state class assigned to each trajectory by visual inspection: regular surface growth (×), irregular surface growth (○), fibrillar growth (□), and bridging network (▽). Shaded regions interpolate between the classified points, and arrows indicate representative simulation snapshots for each regime. One trajectory per point. See also Figures S3 and S4; Tables S1 and S2.

Chain rigidity is introduced independently through harmonic angular potentials. The 25-bead chain follows an $A_5B_5A_5B_5A_5$ architecture, with the two internal B_5 rigid segments occupying beads 6–10 and 16–20. Five consecutive bond angles are constrained around each B_5 segment, with the angular potentials centered on beads 6–10 and 16–20, respectively. Because each bond angle involves the central bead and its two nearest neighbors, the constrained regions extend from beads 5–11 and 15–21, producing two seven-bead rod-like units. All remaining bond angles are left flexible. This construction is intended to isolate the generic physical role of local rigidity in promoting fibrillar growth rather than to reproduce the detailed sequence-specific mechanics of a particular protein. This design separates cohesion from stiffness: a chain can be made more rigid without increasing its attraction strength, and its motif pattern can be rearranged while keeping composition fixed. Rigidity and motif spacing therefore serve as the two sequence-encoded variables in the model. With this minimal generic model, the cohesive bead runs represent β-prone or aggregation-prone motifs at a coarse-grained level, rather than preformed β sheets. The simulations do not resolve residue-level chemistry, strand registry, or hydrogen-bond directionality, the features that generate cross-β order in real amyloid. Instead, the model isolates the physical coupling between motif patterning, chain stiffness, interfacial organization, and non-equilibrium molecular supply from an external bath.

Chains are introduced one at a time into the dilute region at the center of the box between the two interfaces at a prescribed deposition rate $\dot{m}$, which sets the interval between successive insertion events. Each newly introduced chain is assigned an initial velocity biased toward one of the two interfaces, with magnitude controlled by the initial drift velocity $|v_0|$. No external force acts after insertion; therefore, $|v_0|$ represents an initial kinetic bias, or initial kick, rather than a sustained flow field. The two parameters $\dot{m}$ and $|v_0|$ are varied independently of one another and independently of chain architecture, allowing us to separate the effects of material addition, initial approach speed, and molecular design. Figure 2 uses semi-arrested condensate slabs built from the same 25-bead chains that are later introduced into the dilute region. Thus, the interface and the incoming material have identical molecular composition. The interface can move and locally rearrange, but relaxes more slowly than the dilute incoming chains as it is part of a dense, pre-assembled condensate phase. For the controlled growth-regime analysis in Figures 3, 4, and 5, we use a simplified interfacial geometry in which the same chains are arranged into flat surfaces and held fixed. This provides a stationary planar template for growth, isolating the effects of $\dot{m}$, $|v_0|$, chain rigidity, and motif patterning from fluctuations of the dense phase itself. Figures 3 and 4 use two opposing fixed interfaces, whereas Figure 5 uses a single fixed interface.

Rigid segments convert interfacial alignment into directional fibrillar growth

We first tested whether interfacial growth is controlled only by molecular accumulation at the condensate surface, or whether the mechanical properties of chains adsorbed at the interface determine the resulting structure. To isolate this effect, chains were deposited onto a mobile dense-phase interface while varying only the angular stiffness along the backbone. Stiff chains, $K_\theta = 50\ \epsilon$/rad^2, formed anisotropic protrusions that extended away from the condensate surface (Figure 2A). In contrast, chains with the same bead pattern but low angular stiffness, $K_\theta = 0.1\ \epsilon$/rad^2, were adsorbed at the interface but relaxed into laterally spreading, disordered surface aggregates rather than protrusive fibrils (Figure 2B).

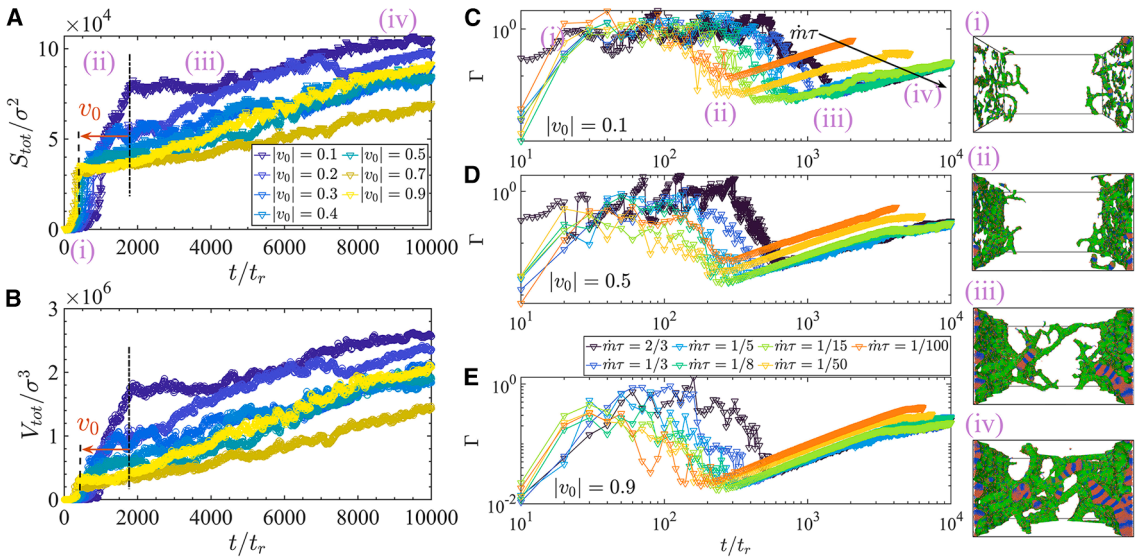


Figure 4. Surface growth under different flux conditions

All figures use stiff chains (K_θ = 50 ϵ/rad^2) on the fixed structured surface, with time measured in insertion intervals per face.

(A) Exposed surface area and (B) enclosed volume of the largest connected cluster at a fixed deposition rate, shown across a range of initial drift velocities. The inset in (B) shows the corresponding volume-loading ratio.

(C–E) Molecular accumulation per unit surface area, $\Gamma(t)$, is defined as the number of chains accumulated up to time t divided by the instantaneous surface area they occupy. Curves are shown for three initial drift velocities for different deposition rates $\dot{m}$. Simulation snapshots at four representative times (i–iv) illustrate the corresponding growth morphologies. The two slowest-supply curves end early because those runs stopped before reaching the intended total number of inserted chains (parameters in Table S1). See also Figures S1, S2, and S6; Table S1.

The morphological difference appears already at early deposition times. For stiff chains, locally aligned segments accumulate at the interface and form orientationally ordered clusters that act as precursors to fibril nucleation. Continued molecular arrival thickens these clusters and promotes ordered extension out of the interfacial plane, while the growing structures remain anchored to the condensate surface (Figure S5). Flexible chains also accumulate at the interface, but their conformational compliance prevents persistent orientational correlations. Local density therefore relaxes laterally along the surface rather than forming stable out-of-plane protrusions. Because these two simulations differ only in angular stiffness, this comparison identifies backbone rigidity as the variable that converts interfacial adsorption into directional fibrillar growth.

We next quantified whether the interface imposes orientational order before fibrillar growth begins. Interfacial conformations in biomolecular condensates depend on distance from the interface, molecular identity, and thermodynamic state.[35,37,38] We therefore measured local alignment directly using the nematic order parameter

$$S(d) = \left\langle \frac{1}{2}\left(3\cos^2\alpha - 1\right)\right\rangle, \qquad \text{(Equation 1)}$$

where α is the angle between a model-specific molecular director and the interface normal, and d is the distance from the interface. Director definitions for each model are provided in STAR Methods. With this convention, isotropic orientations give S = 0, directors lying preferentially in the interfacial plane give S = −1/2, and directors aligned along the interface normal give S = 1.

Both interfacial geometries impose planar alignment on the chain segments that directly contact the surface. At the fixed structured surface, the angle-constrained rod-like units approach the planar limit within the first few bead diameters, $d \lessapprox 3\sigma$, while the shorter linker runs show the same tendency with weaker orientational order (Figure S3). Beyond this contact layer, $S(d)$ exhibits oscillatory structure, indicating that the surface induces local orientational layering rather than a purely monotonic decay. The mobile condensate interface produces the same qualitative bias, but with smaller amplitude and over a broader, diffuse interfacial region (Figure S4). In this case, planar ordering is strongest on the dilute side of the interface, approximately 4σ from the surface, and relaxes toward near-isotropic values inside the dense phase. The measured profiles extend to only $d \approx 8\sigma$ on the dilute side; within this window the planar bias has weakened but has not completely vanished (quantitative values are summarized in Table S2). Beyond the contact layer, the structured surface induces longer-ranged orientational organization. Rather than relaxing monotonically to $S(d)$ = 0, the order parameter alternates with a characteristic spacing of approximately 8σ, with positive maxima near $d \approx 5\sigma$ and again near $d \approx 13\sigma$, before approaching isotropic values at larger distances. As positive S corresponds to alignment along the interface normal, and because this spacing is comparable to the separation between successive rod-like units along an extended chain, these maxima are consistent with outer segments of surface-anchored chains projecting into the dilute region while their leading segments remain adsorbed. The interface therefore generates a layered orientational structure: a

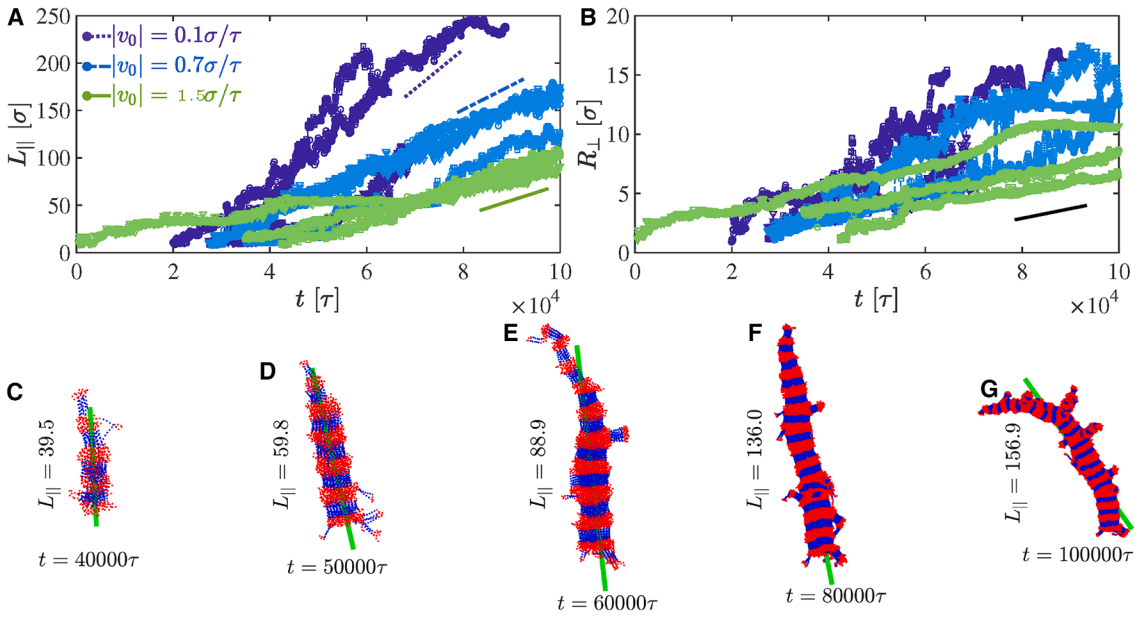


Figure 5. Filament growth kinetics

(A) Longitudinal filament length, $L_{\parallel}$, as a function of time for different initial drift velocities $|v_0|$. The slopes of the approximately linear elongation regimes decrease as $|v_0|$ increases.

(B) Transverse filament extent, $R_{\perp}$, as a function of time, showing a much weaker dependence on $|v_0|$.

(C–G) Representative filament configurations from the simulation snapshots at successive times. The green line indicates the principal filament axis used to define $L_{\parallel}$ and $R_{\perp}$. See also Table S3.

planar contact layer at the surface, followed by segments that preferentially orient away from it. This geometry is precisely the local arrangement required for protrusive elongation. Once a rod-like unit aligns and forms cohesive contacts, subsequently arriving chains encounter a partially ordered interfacial template rather than an isotropic surface.

Rigidity is therefore one control parameter for interfacial morphology, and the molecular arrival rate is a second. By sweeping both variables at the fixed structured surface, we find that the final morphologies organize into four classes in the K_θ and $\dot{m}$ phase space (Figure 3). Flexible chains produce regular or irregular surface growth across the range of rates examined. Increasing backbone stiffness enables protrusive fibrillar growth, and the stiffest chains at the highest rates extend far enough to bridge the two opposing interfaces. This diagram should be interpreted as a nonequilibrium morphology map rather than an equilibrium phase diagram: each state point corresponds to a single trajectory, and the boundaries organize observed outcomes rather than thermodynamic phase transitions. The placement of cohesive motifs along the chain also affects the growth pathway. When cohesive beads are scattered along the backbone at fixed bead composition, with the rigid segments held at the same backbone positions as in the patterned chain, material accumulates primarily as a broad, laterally spreading surface layer (surface-thickened) rather than as protrusive fibrils (Figure 2C). In this comparison, however, we also remove one of the two like-type attractions. The result therefore reflects a combined change in motif placement and cohesive specificity, rather than an isolated test of motif placement alone.

Molecular flux partitions material between spreading and extension

Material reaching a condensate interface can be incorporated in two geometrically distinct ways: it can spread laterally along the surface, increasing interfacial coverage, or it can extend outward into the dilute phase, increasing the height and volume of the growing assembly. We therefore asked how the imposed molecular flux partitions incoming material between these two modes. To quantify this competition, we tracked the exposed area and geometric envelope of the growing structure using the convex hull of the largest adsorbed connected cluster. A connected cluster is defined with a bead-bead distance cutoff of 1.5σ; only molecules belonging to this adsorbed cluster were included in the surface-area and volume measurements. This description treats the aggregate as an evolving interfacial object whose growth statistics reflect both molecular attachment and transport to the interface.[61]

Increasing the initial drift velocity reduces the time required for surface coverage (Figure 4A). At early times (Figures 4i and 4ii), deposited chains accumulate on the two opposing surfaces of the condensates in a spreading-dominated transient, visible as density fronts advancing inward from both interfaces (Figure S6). The cluster area grows rapidly until it reaches a shoulder at $3 \times 10^4\sigma^2$, after approximately 1,000 chain insertions for the lowest initial drift velocity, $|v_0| = 0.1\ \sigma/\tau$, but after only about 300 insertions for the fastest case, $|v_0| = 0.9\ \sigma/\tau$, with a monotonic decrease in this coverage time at intermediate $|v_0|$. The convex-hull volume shows the same trend (Figure 4B), indicating that faster arrival increases not only lateral coverage but

also the rate at which the aggregate envelope expands. After the initial coverage transient, the exposed area grows approximately linearly in time (Figures 4iii and 4iv), consistent with a regime in which newly arriving chains feed both continued interfacial thickening and outward extension. Keeping the deposition rate $\dot{m}$ constant, this linear-growth regime begins earlier as $|v_0|$ increases, showing that the initial drift velocity controls the timescale on which the system crosses from initial spreading to sustained three-dimensional growth.

To separate lateral spreading from outward growth, we followed the loading ratio $\Gamma(t) = \frac{N_{ins}(t)}{A_{hull}(t)}$, where $N_{ins}(t)$ is the number of chains deposited up to time t and $A_{hull}(t)$ is the exposed area of the largest connected cluster (Figures 4C–4E; see also Figure S1). Thus, $\Gamma(t)$ measures molecular accumulation per unit exposed area. In all cases, $\Gamma(t)$ first increases as deposited chains are captured by a compact interfacial aggregate. It then decreases by one to two orders of magnitude as the aggregate spreads laterally faster than new material is added per unit area. At later times, $\Gamma(t)$ recovers and approaches a slowly varying regime in which deposition and surface expansion proceed together. This is the regime in which sustained fibril growth occurs. Changing the deposition rate $\dot{m}$ primarily shifts the timing of this dip-and-recovery process, with the strongest rate dependence observed at the lowest initial drift velocity. At long times, curves obtained at different deposition rates converge to within approximately a factor of two for each initial drift velocity. This convergence is consistent with the kinematic identity derived in STAR Methods: once both $N_{ins}(t)$ and $A_{hull}(t)$ grow approximately linearly in time, their ratio approaches a rate-insensitive asymptotic scale. The deposition rate and initial drift velocity therefore act together to select the growth trajectory. The end states across the joint deposition-drift grid show this coupling directly (Figure S2): frequent deposition favors protrusive growth when the initial drift is weak, whereas large initial drift velocities rapidly cover the surface and bias the system toward lateral spreading, thereby suppressing outward extension. Aging in this model is therefore not a single pathway. It is a family of nonequilibrium trajectories selected by how material is delivered to, and captured by, the condensate interface.

Faster initial arrival slows filament elongation

Once a protrusion has nucleated, the relevant question shifts from whether material is captured at the interface to how efficiently the captured material is converted into axial growth. Deposition initially breaks interfacial symmetry at a small number of aligned clusters. These clusters then develop into protrusions that elongate into the dilute region while also thickening laterally. To separate axial extension from transverse growth, we measured two geometric observables for each growing filament: the length along its principal axis, $L_{\parallel}$ (Figure 5A), and the transverse extent, $R_{\perp}$ (Figure 5B). We obtained both quantities from the gyration tensor of the connected growing assembly (STAR Methods). Both $L_{\parallel}$ and $R_{\perp}$ increase approximately linearly during the growth regime under continued molecular deposition (Figures 5A and 5B). However, the longitudinal elongation rate decreases systematically with increasing initial drift velocity $|v_0|$, whereas the transverse extent depends only weakly on $|v_0|$. Thus, a stronger initial drift does not simply enhance filament growth. Instead, faster arrival promotes rapid interfacial capture and surface coverage, but appears to reduce the efficiency with which incoming chains are incorporated into the protruding filament axis. Filament elongation is therefore not determined by molecular deposition alone, but by how deposited chains are orientationally incorporated into the existing protrusion after interfacial nucleation has occurred.

Averaged over three trajectories per condition, the mean rates of both axial elongation and transverse thickening decrease monotonically as the initial drift velocity increases (Table S3). One explanation is where incoming chains are incorporated. Chains that attach along the lateral surface of an existing fibril or protrusion mainly increase its radial extent $R_{\perp}$, whereas chains captured near the filament tip contribute more directly to longitudinal elongation $L_{\parallel}$. Larger $|v_0|$ promotes rapid interfacial capture and surface coverage, but appears to reduce the efficiency with which deposited chains are incorporated into the growing protrusion. We therefore interpret the data as a qualitative slowdown of filament growth with increasing initial drift velocity, rather than assigning a precise rate-reduction factor. The growing structures remain strongly anisotropic throughout, with one gyration-tensor eigenvalue much larger than the other two, $\lambda_1 \gg \lambda_2 \approx \lambda_3$. Thus, the symmetry breaking established during interfacial nucleation persists during subsequent growth. The slowdown at larger $|v_0|$ suggests a competition between rapid surface coverage and efficient incorporation at the protruding filament, although separating these contributions would require independently resolving arrival, docking, and surface-diffusion kinetics.

DISCUSSION

Aging of condensates and formation of amyloid fibrils are complex non-equilibrium processes which complicate dissection of the role of molecular sequence and environmental contributions. Recent experiments show that condensate interfaces play an active role in fibril formation, where monomer availability is often gated by dynamic condensates, thereby providing direct roles for condensates in influencing amyloid formation. In this work, we designed FD-MD, a non-equilibrium coarse-grained framework in which chain architecture, chain rigidity, and molecular transport can be varied independently in the presence of condensate interfaces.

Using FD-MD simulations, we show that fibrillization during condensate aging is governed by the interplay between molecular architecture that dictates thermodynamic driving forces and the rate at which chains become available. Specifically, we show that the more rigid β sheet motifs provide the structural capacity for directional fibril growth, while deposition-driven flux controls whether incoming material spreads along the interface or extends away from it. Within this minimal framework, β-prone rigid segments and non-equilibrium molecular deposition are sufficient to generate distinct fibrillar aging pathways at condensate interfaces, as discussed in the literature.[8,9,22,31] The chain rigidity determines whether interfacial adsorption can be converted into directional growth, while the deposition rate determines how incoming material is partitioned between lateral spreading and outward extension. Thus, chains with identical

molecular architecture can age along distinct morphological routes simply because the rate and manner of molecular delivery change. In this sense, transport is not a passive background condition but an independent control variable. The resulting morphologies suggest that some diversity in condensate-aging phenotypes across protein systems may reflect movement through a low-dimensional space of rigidity, interfacial ordering, and molecular deposition, rather than requiring fundamentally distinct mechanisms in each case. Within this framework, the model connects naturally to interface-associated fibrillation in hnRNPA1 condensates,[9,19] to suppression or spatial redistribution of fibrillation in multicomponent condensates,[8,21] and to systems in which amyloid formation occurs more broadly throughout condensate interiors, such as Tau.[22]

The structural basis for directional growth is the orientational field imposed by the interface. In our simulations, condensate interfaces are not passive collecting surfaces. They organize arriving chains before large-scale assembly begins, producing a planar contact layer at both the fixed structured surface and the mobile but semi-arrested condensate interface. At the structured surface, this contact layer is followed by segments preferentially oriented along the interface normal, creating a layered architecture that is already biased toward protrusive elongation. This organization provides a local template for fibril growth: chains adsorb in the interfacial plane, while segments extending away from the surface define the direction along which protrusions can advance. This behavior places the deposited chains closer to the adsorbent class of interfacial orientation reported in multicomponent condensates, rather than the radially extended conformations associated with scaffold chains.[35,38] It also explains why backbone rigidity is essential. Only chains stiff enough to preserve extended, orientationally correlated conformations can exploit the order imposed by the interface and convert it into persistent out-of-plane growth.

The distinct growth regimes also generate mesoscale observables that can be compared with experiments or higher-resolution simulations. Convex-hull geometry separates lateral surface coverage from anisotropic extension, the molecular loading ratio $\Gamma(t)$ reports how accumulated material is distributed over the exposed aggregate area, and the gyration-tensor spectrum, $\lambda_1 \gg \lambda_2 \approx \lambda_3$, identifies persistent symmetry breaking along the filament axis. These quantities do not require residue-level resolution and therefore bridge coarse-grained simulations, microscopy-based morphology measurements, and continuum descriptions of interfacial growth. In the present model, the fixed-surface protocol isolates this transport-assembly coupling by suppressing competing bulk aggregation and dense-phase rearrangement. Together with related nonequilibrium simulations of molecular transport through condensates,[44] our results support imposed-deposition models as a controlled route for dissecting how material flux, interfacial organization, and molecular architecture jointly determine fibrillation during condensate aging.

Taken together, these results separate interfacial aging into two coupled but distinguishable axes. Sequence-encoded rigidity determines whether a chain is structurally competent to sustain directional fibrillar growth, whereas molecular deposition selects which morphological pathway that competence follows. Both axes are experimentally tunable in principle: sequence or chemical modification can alter rigidity and motif patterning, while expression level, exchange kinetics, transport, or remodeling can alter the rate at which material reaches condensate interfaces. These predictions can be tested by varying molecular delivery at fixed sequence and solution conditions, and measuring surface coverage, fibril aspect ratio, and interfacial localization of new material. If this separation holds in cellular condensates, it suggests a practical possibility: aging trajectories could be redirected by modulating molecular delivery, without altering the protein sequence itself. More broadly, the framework connects sequence-encoded molecular structure to driven mesoscale organization, and should apply beyond amyloid to condensates whose aging is shaped by both what their components encode and how rapidly those components arrive.

Limitations of the study

The FD-MD coarse-grained model is designed to isolate how molecular delivery couples to interfacial organization during condensate aging. In this minimal representation, rigid segments along the chain mimic β-prone regions that can sustain extended, anisotropic conformations, while attractive bead blocks represent generic aggregation-prone motifs. This abstraction allows rigidity, cohesion, motif placement, and deposition kinetics to be varied independently, but it also defines the limits of the model.

We used a fixed structured surface for the main growth-regime analysis because it provides a stationary interface normal and a reproducible template for quantifying spreading and protrusive growth. This choice removes condensate curvature, capillary fluctuations, and material exchange with a deformable dense phase. The semi-arrested condensate interface state is not imposed by an external viscous drag; rather, it emerges from the slow relaxation of the dense rigid-flexible block copolymer molecular assembly, providing a coarse-grained analog of glassy or crust-like condensate surfaces that can develop during aging.[20] These simulations show that the same qualitative interfacial orientational bias persists at a mobile condensate interface, although a fully matched model including condensate deformation, exchange, and fibril growth remains a natural next step. The molecular interactions are also intentionally generic. Isotropic Lennard-Jones attractions capture nonspecific cohesive interactions between aggregation-prone segments, but they do not resolve hydrogen-bond directionality, strand registry, or cooperative cross-β sheet formation. Thus, the fibrils in this model should not be interpreted as residue-resolved amyloid structures. Instead, they are coarse-grained fibrillar assemblies stabilized by cohesive attraction and chain rigidity. In this sense, the rigid segments act as pre-encoded structural elements that allow surface-bound chains to maintain alignment, bundle, and grow into long anisotropic protrusions. We also did not define an operational critical nucleus. We therefore do not report critical nucleus sizes, nucleation barriers, nucleation rates, or rate enhancements. Future extensions should include curved and fluctuating interfaces, explicit directional interactions,[51,52] matched bulk and interfacial simulations, and controlled seeding analyses in which pre-formed aligned clusters are inserted either at the interface or in the dilute phase.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Davit A. Potoyan (potoyan@iastate.edu).

Materials availability

This study did not generate new materials.

Data and code availability

- All data reported in this study will be shared by the lead contact upon request.
- All original code has been deposited at Zenodo and is publicly available as of the date of publication. The DOI is listed in the key resources table.
- Any additional information required to reanalyze the data reported in this study is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.B. and D.A.P.; methodology, S.B.; software, S.B.; investigation, S.B.; formal analysis, S.B.; visualization, S.B.; writing – original draft, S.B. and D.A.P.; writing – review and editing, S.B. and D.A.P.; supervision, D.A.P.; funding acquisition, D.A.P.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.isci.2026.117855.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
LAMMPS (3 Mar 2020)	Thompson et al.[62]	https://www.lammps.org
OVITO 3.11	Stukowski et al.[63]	https://www.ovito.org
Python 3.10	Python Software Foundation	https://www.python.org; RRID: SCR_008394
SciPy 1.17	Virtanen et al.[64]	https://scipy.org; RRID: SCR_008058
FD-MD analysis code (Python and MATLAB scripts)	This paper	Zenodo: https://doi.org/10.5281/zenodo.22904932

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Omitted as our study does not involve biological models.

METHOD DETAILS

Molecular model of deposited chains

The deposited chains are identical to the chains that form the pre-existing condensate interface. This choice mimics an experimental situation in which the same aggregation-prone molecular species is continuously delivered to an already formed condensate. The distinction between "interface" and "deposited material" is therefore not compositional, but dynamical: interfacial chains begin in a dense, semi-arrested state, whereas deposited chains are introduced into the dilute region at controlled time intervals and subsequently adsorb, rearrange, or assemble at the condensate surface. The chains are represented as coarse-grained bead–spring polymers with two independently specified features: five-bead β-prone runs that carry the angular constraint, and flexible five-bead linker runs between them. Both bead types attract their own kind, so it is the angular constraint rather than the attraction that distinguishes the two. The architecture is generic: bead types stand for classes of residue rather than for particular ones. The β-prone runs represent aggregation-prone motifs, not preformed β sheets, and the model does not resolve residue-level strand registry.

Fibril-forming biomolecular chains are represented as beads connected by finitely extensible nonlinear elastic (FENE) bonds with potential

$$U_{\mathrm{FENE}}(r) = -\frac{1}{2}kR_0^2 \ln\left(1 - \frac{r^2}{R_0^2}\right),$$

where r is the distance between adjacent beads. The fixed structured-surface simulations use $k = 50\ \epsilon/\sigma^2$, whereas the separate mobile-slab architecture comparison in Figure 2 uses $k = 30\ \epsilon/\sigma^2$. Both use $R_0 = 1.5\sigma$, with bead diameter σ defining the unit of length.

Chain rigidity is imposed via a harmonic angular potential,

$$U_\theta = K_\theta(\theta - \theta_0)^2,$$

where θ is the angle between consecutive bonds and $\theta_0 = 180^\circ$. In the 25-bead production chain, angular potentials are applied only within two internal segments, spanning bead indices 6–10 and 16–20. These constraints generate two rod-like units associated with the β-prone blocks. The rigid units are separated by fully flexible linker segments composed of the other bead type; these linker segments do not carry an additional angular potential. The principal extended-chain simulations use $K_\theta = 50\ \epsilon/\mathrm{rad}^2$; the matched flexible comparison in Figure 2B uses $K_\theta = 0.1\ \epsilon/\mathrm{rad}^2$; and the morphology map uses $K_\theta = 0.01$, 0.1, 1, 2, 5, and 50 ϵ/rad^2 (see Figure 3). We report K_θ directly rather than assigning a measured persistence length of the rigid blocks.

Nonbonded interactions are modeled using a Lennard-Jones (LJ) potential,

$$U_{\mathrm{LJ}}(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right].$$

In the fixed structured-surface runs and Figures 2A and 2B, like bead types interact attractively with $\epsilon = 1$ and cutoff $r_c = 3\sigma$, while unlike types use the repulsive Weeks–Chandler–Andersen cutoff $r_c = 2^{1/6}\sigma$. Following the Kremer– Grest convention, we exclude bonded first neighbors from the nonbonded interaction. In the exploratory randomized run in Figure 2C, only the red–red pair is

attractive; blue–blue and unlike pairs use the WCA cutoff. Consequently, Panel C changes both bead placement and the interaction matrix.

Like-type Lennard–Jones attractions provide cohesion, while the angular potential independently favors extended conformations across each β-prone run and its flanking linker beads. Together these terms allow elongated aggregates to form without resolving backbone hydrogen-bond directionality, side-chain packing, strand registry, or cooperative sheet formation. Related reduced-interaction models have been used to study generic amyloid assembly[47,50–52]; the present model resolves mesoscale morphology rather than residue-level polymorphism.

Simulation protocol

Molecular dynamics simulations were performed using LAMMPS,[62] and simulation snapshots were rendered with OVITO.[63] All 25-bead-chain simulations were carried out in reduced Lennard–Jones units, with $\epsilon = 1$, $\sigma = 1$, particle mass $m = 1$, and $\tau = \sqrt{m\sigma^2/\varepsilon}$. Equations of motion were integrated using the velocity-Verlet algorithm with timestep 0.01 τ. Unless otherwise stated, simulations were performed at reduced temperature $T^* = 1.0$ using a Nosé–Hoover thermostat with damping time 1.0 τ. For deposition runs in which newly inserted chains were assigned an initial drift velocity, the deposited chains were subsequently propagated under NVE dynamics. This choice preserves the imposed initial velocity bias after insertion; applying a thermostat directly to these chains would relax the imposed drift and mix it with thermal velocity fluctuations. In the mobile-slab simulations shown in Figures 2 and 3, the thermostat was applied to all beads, allowing the dense phase and deposited chains to exchange energy while the condensate interface remained dynamically responsive. We also tested a highly arrested-interface limit by suppressing motion of the pre-existing interfacial beads while keeping the same molecular architecture and interaction parameters. This protocol mimics the limiting case of a strongly viscoelastic or crust-like condensate interface. The resulting morphology map was not qualitatively changed (while increasing the simulation runtime), indicating that the main fibril-growth regimes are set primarily by chain rigidity and deposition protocol rather than by residual mobility of the interfacial template.

Three model classes were analyzed, with the structured-surface class implemented in two geometries. The mobile 25-bead-chain slab in Figure 2, the two-faced and one-sided fixed structured surfaces in Figures 3, 4, and 5, and the separate 200-bead sticker–spacer slabs in Figure S4 differ in architecture, interactions, mobility, interface structure, and driving. Comparisons across model classes or structured-surface geometries are qualitative unless stated otherwise.

Mobile 25-bead-chain slab for the architecture comparison

Figure 2 was initialized from 8,000 fully mobile 25-bead chains (200,000 beads) in a periodic $200\sigma \times 120\sigma \times 120\sigma$ box. One chain insertion was attempted without a prescribed initial velocity every 500 steps in a central region, until 20,000 chains had been introduced. A weak stationary x-directed field derived from

$$U(x) = U_0\left[\tan^{-1}\left(\frac{x + x_0}{\delta}\right) - \tan^{-1}\left(\frac{x - x_0}{\delta}\right)\right] \quad \text{(Equation 2)}$$

acted on all beads throughout deposition, with $U_0 = 0.0015\epsilon$, $x_0 = 20\sigma$, and $\delta = 5\sigma$. The initial slab remained mobile and was recentered along x; it was not a fixed substrate. Panels A and B use the patterned chain at $K_\theta = 50$ and 0.1, respectively. Panel C uses $K_\theta = 50$ and randomized bead placement at fixed composition, together with the interaction-matrix change described above. Each panel is one trajectory.

Two-faced structured planar surface

The morphology map and global surface-growth analysis (Figures 3 and 4) use a one-bead-thick structured surface built from 558 of the same 25-bead chains (13,950 beads), laid flat in the plane and held fixed. The surface therefore presents the same bead types, and the same cohesive interactions, as the chains arriving at it. Interactions among surface chains are switched off, so the surface acts as a rigid template rather than as a cohesive phase. The simulation box is periodic in all directions with $L_x = L_y = 120\sigma$ and $L_z = 200\sigma$, and the surface normal lies along z. Periodicity makes the single surface present two faces to the intervening dilute region. This setup provides a stationary interfacial plane but excludes capillary fluctuations and exchange with a dense phase. Two insertion regions attempt one chain toward each face every τ_d integration steps, with an intended total of 20,000 chains. A run with $\tau_d = 300$ therefore spans 3×10^6 integration steps. All runs with $\tau_d \leq 1500$ reached the intended total. The $\tau_d = 5000$ and $\tau_d = 10000$ runs were stopped earlier, between 21% and 78% of the intended duration depending on condition, so they sample a correspondingly smaller cumulative chain count. We therefore report them over the interval actually simulated and do not compare their end states to those of the completed runs.

One-sided structured planar surface

The filament-length and radial-extent analysis in Figure 5 uses a separate one-sided growth box with a fixed surface of 1,008 such chains (25,200 beads), periodic x and y boundaries, nonperiodic reflecting boundaries along z, and lateral dimensions of approximately $170\sigma \times 170\sigma$. These runs use $K_\theta = 50$, an insertion interval of 500 steps, and $|v_0| = 0.1$, 0.7, or 1.5. A total of 20,000 chains is introduced toward the single surface, with three trajectories analyzed for each $|v_0|$. This geometry was used only for the elongation analysis and should not be conflated with the periodic two-faced setup.

Explicit mobile condensate slab (qualitative orientational comparison)

We separately analyzed fully mobile condensate slabs containing up to 2×10^5 beads. These slabs are formed by liquid–liquid phase separation of 200-bead, 11-type sticker–spacer chains using the model of Biswas and Potoyan[53]; their architecture and interactions differ from the 25-bead deposition model. The slabs span the periodic x boundary and present two diffuse interfaces of width $\approx 20\sigma$. No FD-MD initial-velocity bias is assigned, but the analyzed trajectories include continuous molecular addition and the externally applied x-directed fields of the original mobile-slab protocol. We use these simulations only to compare the directional sense and spatial extent of local orientational bias (Figure S4), not as an equilibrium control or as the undriven limit of the structured-surface model.

Flux implementation

To probe how imposed molecular flux affects assembly at the structured surface, we use a thermostatted protocol in which polymer-chain insertions are attempted at controlled intervals. The particle number increases throughout each run, so this is an open, nonequilibrium setup at constant volume and thermostat temperature. Without an initial directional bias, chains can form aggregates in the intervening dilute region before reaching the surface. The biased protocol reduces this competing pathway but does not provide a matched solution-versus-interface nucleation comparison.

The protocol increases the frequency of surface encounters while suppressing aggregation in the intervening dilute region. One insertion command acts in each half of that region. For each chain, the two velocity components parallel to the surface are drawn uniformly from $[-v_0, v_0]$, while the normal component is drawn from the corresponding interval directed toward the nearer surface face. Thus, v_0 bounds each Cartesian component, the mean normal component has magnitude $v_0/2$, and the total initial velocity can exceed v_0. Insertions that overlap existing material are rejected.

We treat the deposition rate and the initial drift velocity as independent protocol inputs. The symbol τ_d denotes the interval in integration steps between insertion attempts in each half-box. With timestep $\Delta t = 0.01\tau$, the nominal per-face deposition rate is $1/(\tau_d \Delta t)$ and the total nominal rate from both insertion regions is $2/(\tau_d \Delta t)$, before rejected attempts are accounted for. The morphology map uses the dimensionless nominal per-step attempt frequency $\dot{m} = 1/\tau_d$ as its horizontal coordinate. After insertion, chains evolve under thermostatted equations of motion with no sustained external force, so the imposed velocity is an initial condition rather than a sustained drift. Two considerations set the scale over which that initial condition persists, and both matter for interpreting the initial drift velocity dependence reported below.

First, the appropriate thermal reference is the center-of-mass velocity of a whole 25-bead chain, $\sqrt{k_B T/M} = 0.2\,\sigma/\tau$, rather than the single-bead value of $1\,\sigma/\tau$. The mean imposed normal drift is $v_0/2$, which ranges from $0.05\,\sigma/\tau$ at $|v_0| = 0.1$, below the thermal center-of-mass scale, to $0.45\,\sigma/\tau$ at $|v_0| = 0.9$ and $0.75\,\sigma/\tau$ at $|v_0| = 1.5$, several times above it. The initial drift velocities sampled here therefore span the crossover from diffusion-dominated to directed arrival, which is the range over which a dependence on $|v_0|$ is expected.

Second, the Nosé–Hoover friction coefficient is a single global feedback variable that fluctuates about zero once the target temperature is reached. Unlike a Langevin friction, it applies no systematic drag to an individual chain, and in the dilute gap between the two faces there are few collisions to randomize a tagged chain's center-of-mass motion. The damping parameter of 1τ thus sets the response time of the thermostat and should not be read as a decorrelation time for the imposed drift; we therefore do not equate the two. We did not measure the velocity autocorrelation or the insertion-to-contact transit time independently, and a direct measurement of the latter would be the sharpest test of this interpretation. Quantitative comparison with protein expression, flow, or chemical-potential gradients requires an explicit parameter mapping. Biased insertion reduces aggregation in the dilute region and thereby isolates driven growth at the structured surface. This deliberate exclusion of a competing pathway limits the conclusions to the imposed protocol.

QUANTIFICATION AND STATISTICAL ANALYSIS

Surface growth analysis

For the structured-surface trajectories, coordinates were first unwrapped across the periodic z boundary so that material associated with both surface faces occupied one continuous coordinate representation. Connected components were constructed from all deposited beads using a distance cutoff of 1.5σ, and the largest component was used for the global shape analysis. Its convex-hull area $A_{\mathrm{hull}}(t)$ and convex-hull volume $V_{\mathrm{hull}}(t)$ were calculated with the ConvexHull implementation in SciPy.[64] These quantities are geometric envelope measures; in particular, A_{hull} does not resolve concave internal surfaces. The Python and MATLAB scripts for this and the following analyses are deposited at Zenodo.[65] An AI-assisted tool was used to check these scripts for correctness.

The plotted quantity $\Gamma(t)$ is a surface-loading proxy,

$$\Gamma(t) = \frac{N_{\mathrm{ins}}(t)}{A_{\mathrm{hull}}(t)},$$

where $N_{\mathrm{ins}}(t) = 2t/(\tau_d \Delta t)$ is the nominal cumulative number of introduced chains from both insertion regions, evaluated at the elapsed time of each frame rather than assumed to reach the intended total. It is not a direct local number density because the numerator

includes all inserted chains. The ratio decreases whenever convex-hull area grows faster than the introduced-chain count and increases when it grows more slowly.

Orientational order was quantified using Equation 1. In the structured-surface model, directors were calculated separately for the end-to-end vectors of the three five-bead linker runs and for the two seven-bead angle-constrained units. Distance d was measured from the nearest surface face, and data from both faces were pooled. In the mobile condensate model, directors were five-bead local tangents restricted to the stiff sticker-core region. The two interface positions were recalculated in every frame from the half-maximum crossings of the bead-density profile; signed distance is negative in the dense phase and positive in the dilute phase. The mobile-condensate profiles contain 32 frames for the 2×10^5-bead slab and 13 frames for the 5×10^4-bead slab, each from a single trajectory.

Kinematic interpretation of the surface-loading proxy

The logarithmic slope of the surface-loading proxy $\Gamma(t)$ satisfies the identity

$$\frac{d \ln \Gamma}{d \ln t} = \frac{d \ln N_{\mathrm{ins}}}{d \ln t} - \frac{d \ln A_{\mathrm{hull}}}{d \ln t}. \qquad \text{(Equation 3)}$$

For approximately constant successful insertion, $N_{\mathrm{ins}} \propto t$, so the slope becomes $1 - d \ln A_{\mathrm{hull}}/d \ln t$. If both quantities become linear at late times, their ratio approaches a constant. This algebraic interpretation explains the non-monotonic trend without assigning either extremum to an operational nucleation event.

Figure S1 shows the corresponding curves at the single completed interval $\tau_d = 300$, and representative end states across the flux grid are collected in Figures S2 and S6 resolves the same grid in space and time, as bead-density profiles across the slab, and shows the two accumulating fronts meeting near the midplane in the bridging corner. Figure S5 reports the center-of-mass mean squared displacement of the deposited assembly, which crosses over from an early superdiffusive stage to an arrested plateau; we use it only to show that the anchored assembly stops translating and draw no quantitative transport conclusion from it.

Interpretation of the orientational order parameter

Because $S(d)$ is only the second moment of the orientational distribution, it does not determine the full angular distribution. We therefore do not convert it into an orientational entropy, free-energy barrier, or nucleation-rate ratio. Figures S3 and S4 show the profiles, and Table S2 collects the measured values.

The angle-constrained units reach $S \approx -0.50$, the planar limit, at all three $|v_0|$ values. This near-saturation is largely a steric consequence of rod-like units meeting a flat impenetrable plane and should not be read as a measure of the orientational bias expected at a deformable condensate interface; the mobile-slab profiles in Figure S4B provide the less constrained comparison. $S(d)$ is also not monotonic beyond the contact layer. It alternates with a period of approximately 8σ, so that material displaced from the surface by roughly one rod length is oriented along the interface normal rather than isotropically. The envelope decays slowly, and $|S|$ falls below 0.05 only beyond $d \approx 40\sigma$. This alternation is specific to the structured surface: in the mobile condensate slabs $S(d)$ remains negative throughout the dilute side, passing through minima of -0.21 and -0.39 in the larger and smaller slabs respectively at $d \approx 4\sigma$. These profiles extend to $d \approx 8\sigma$ on the dilute side and therefore do not resolve the return to isotropic values there.

The accompanying bead-density profile in Figure S3C shows interfacial enrichment that increases with $|v_0|$. Because it depends on the imposed flux, it reflects driven accumulation under the protocol and not an equilibrium adsorption free energy, which would require an undriven reference simulation that the present archive does not contain.

Filament-shape and reproducibility analyses

For Figure 5, deposited beads were joined into connected components when any bead pair lay within 1.5σ, and the largest component was tracked as the filament assembly. Its shape was characterized by the gyration tensor

$$\mathbf{G} = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}}) \otimes (\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}}), \qquad \text{(Equation 4)}$$

where N is the number of beads, $\mathbf{r}_i$ are bead positions, and $\mathbf{r}_{\mathrm{cm}}$ is the center of mass. Diagonalizing $\mathbf{G}$ gives eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ with eigenvectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$, and $\mathbf{e}_1$ defines the filament axis. The axial length $L_{\parallel}(t)$ is the range of bead projections along $\mathbf{e}_1$, and the transverse extent is $R_{\perp}(t) = \sqrt{\lambda_2(t) + \lambda_3(t)}$, the root-mean-square perpendicular distance from that axis. Three trajectories were analyzed for each $|v_0|$ in this one-sided geometry; no operational nucleus or fibril-size threshold was imposed. Because the nine trajectories span different time ranges, no common fit window contains them all, so each trace was fit in full by ordinary least squares and the three slopes at each $|v_0|$ were averaged. The resulting values are given in Table S3. The means decrease monotonically in both directions as $|v_0|$ increases, but individual trajectories overlap between conditions and the scatter at $|v_0| = 0.1$ is comparable to the difference between conditions, so these values establish the direction of the effect and not its magnitude.

Each combination of K_θ and τ_d in the morphology map corresponds to a single trajectory. The map and global kinetic curves are therefore descriptive, the interpolated boundaries are approximate, and run-to-run variability cannot be estimated. As noted above, the two longest insertion intervals correspond to runs stopped before the intended chain total, and we do not draw conclusions that

require comparing their end states with those of the completed runs. The structured-surface orientational profiles pool 100 frames for $|v_0| = 0.1$ and 50 frames each for $|v_0| = 0.5$ and 0.7, again from one trajectory per condition. No finite-size scaling was performed for bridging in either interface geometry.

Provenance of the archived figure panels

Two panels in the main text derive from archived composites whose full provenance is not recoverable. The configuration sequence in Figures 5C–5G is taken from an archived composite that preserves the displayed times but not the condition and seed identifiers of the individual snapshots. Those panels are therefore illustrative only, and every quantitative statement about filament growth uses the nine source-identified traces instead. The morphology map in Figure 3 classifies each end state visually. The four labels summarize morphology rather than a computed order parameter, each point is a single trajectory, and the shaded background interpolates between classified points rather than locating a thermodynamic boundary. Bridging in the highest-supply, highest-stiffness corner depends on the finite separation between the two periodic faces, which we did not test by finite-size scaling.